Cricket Player Performance Tracking Application : CRICKSTATS

Project Scope :

As a Salesforce Admin, I want to create a comprehensive player performance tracking application within Salesforce to capture and analyze detailed match-by-match information of cricket players. This application will include custom objects such as Player and Match Performance with carefully defined fields and relationships to ensure data integrity and accessibility. By leveraging master-detail and lookup relationships, the application will enable robust tracking of individual and team performances, facilitate insightful reporting and analytics, and support the overall management of player statistics, thereby enhancing strategic decision-making and player development initiatives.

1. INTRODUCTION

1.1 Project Overview

CRICKSTATS is a Salesforce-based cricket performance management application built to record, monitor, and evaluate player performances match by match. The system uses custom objects, master-detail and lookup relationships, roll-up summary fields, automation flows, and interactive dashboards to deliver structured and reliable performance insights.

It is designed for coaches, team managers, analysts, and Salesforce administrators who need accurate statistics, real-time analytics, and automated data handling to support team selection, player development, and strategic planning.

1.2 Purpose

The primary objectives of CRICKSTATS are:

- To replace manual record-keeping with a centralized digital system
- To improve the accuracy and accessibility of cricket statistics
- To provide real-time performance analytics through dashboards and reports
- To support player development and data-driven team planning

- To automate data management using Salesforce Flows and validation rules

2. IDEATION PHASE

2.1 Problem Statement

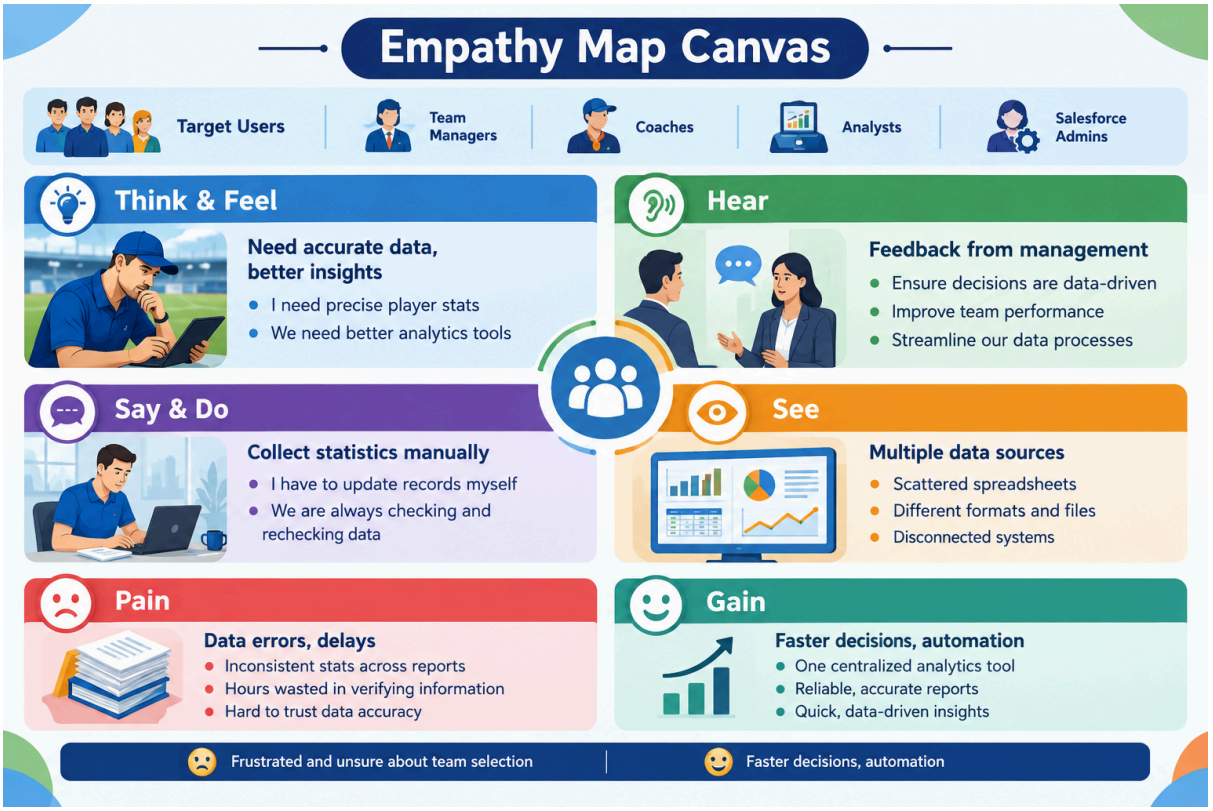
PS ID	Customer	Trying To	But	Because	Feels
PS-1	Cricket team coach	Track player performance across matches	Cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection
PS-2	Team manager	Maintain complete player records	Takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient
PS-3	Sports analyst	Analyze player trends and generate reports	Faces difficulty in comparing performances	No centralized analytics system exists	Confused and dissatisfied



2.2 Empathy Map Canvas

Target Users: Team Managers, Coaches, Analysts, Salesforce Admins

Aspect	Description
Think & Feel	Need accurate data and better insights
Hear	Feedback from management and team staff
See	Multiple scattered data sources
Say & Do	Collect statistics manually
Pain	Data errors, delays, and duplication
Gain	Faster decisions, automation, and clarity



2.3 Brainstorming

Brainstorming is a process of generating and considering multiple possible solutions before choosing a direction for the project. During the CRICKSTATS ideation phase, several application concepts were considered.

Ideas Generated:

- Web-based player management system
- Mobile statistics tracker
- Salesforce-integrated CRM
- AI-based performance prediction
- Dashboard-based analytics

Selected Idea: Salesforce-based performance tracking platform (CRICKSTATS)

2.4 Selected Idea

The selected concept was a Salesforce-based cricket performance tracking platform, named CRICKSTATS. The project material identifies this as the chosen idea following brainstorming.

The selected concept provides a foundation for organizing player and match performance information in Salesforce. The project's later design uses custom objects, fields, relationships, automation, and reporting components to support the intended tracking and analysis activities.



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

1. Login
2. Enter Player Data
3. Record Match Performance
4. Generate Reports
5. Analyze Results
6. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Player Registration
FR-2	Match Performance Entry
FR-3	Dashboard Generation
FR-4	Report Creation
FR-5	Flow Automation

Non-Functional Requirements

NFR-1: Usability
The system must be simple enough for coaches, managers, analysts, and admins to use without technical training.

- Intuitive user interface
- Clear navigation menus
- Easy-to-read dashboards
- Minimal learning curve

NFR-2: Security
Sensitive player data must be protected from unauthorized access.

- Role-based access control
- Secure authentication
- Encrypted data storage
- Regular backups

NFR-3: Reliability
The system must function correctly and store data accurately.

- No data loss
- Automatic backups
- Error handling
- System stability

NFR-4: Performance
The system must respond quickly and handle large data volumes efficiently.

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal response time

NFR-5: Availability
The system must be accessible whenever needed.

- 24/7 availability
- Minimal downtime
- Cloud-based access
- Disaster recovery

NFR-6: Scalability
The system must support growing users, players, and records.

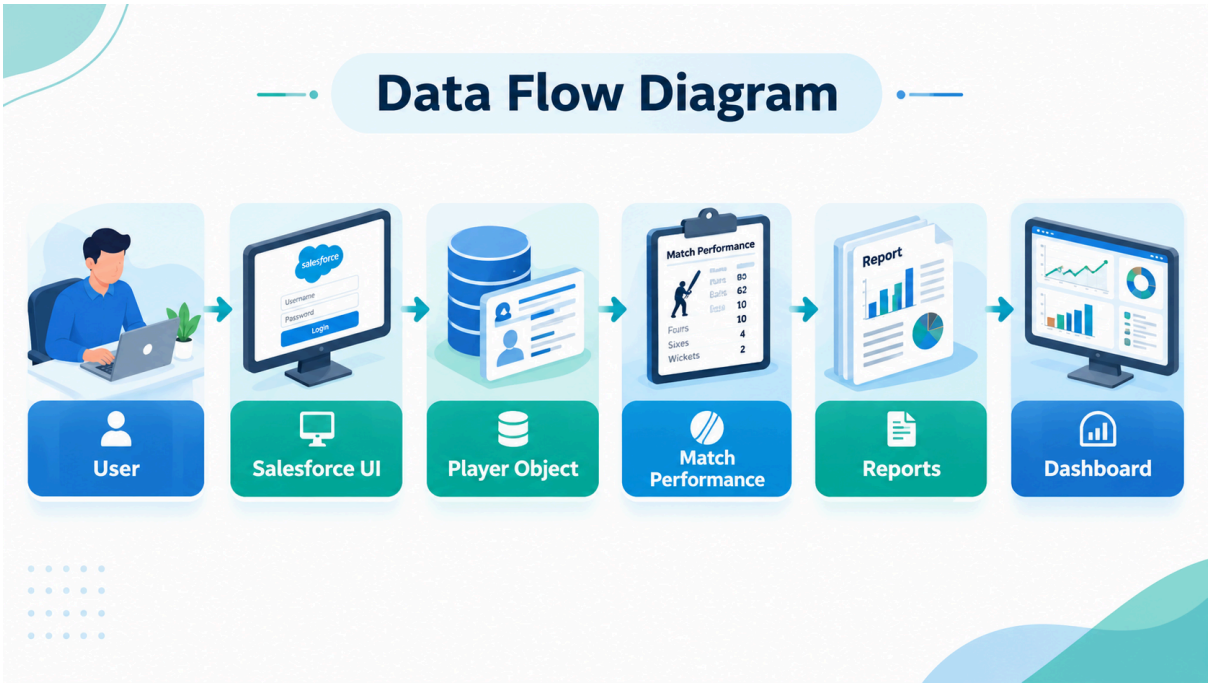
- Handle growing datasets
- Support more teams and leagues
- Expand storage capacity
- Maintain performance under load

3.3 Data Flow Diagram

Data Flow:
User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard

What is a Data Flow Diagram?

A Data Flow Diagram (DFD) is a visual representation that shows how data moves through a system — where it comes from, how it is processed, where it is stored, and where it finally goes. In the CRICKSTATS application, the DFD helps me understand the complete journey of cricket data from the moment a user enters it until it appears on a dashboard.



3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Analytics	Reports & Dashboards
Automation	Salesforce Flows
Backend	Salesforce Objects
Database	Salesforce Cloud

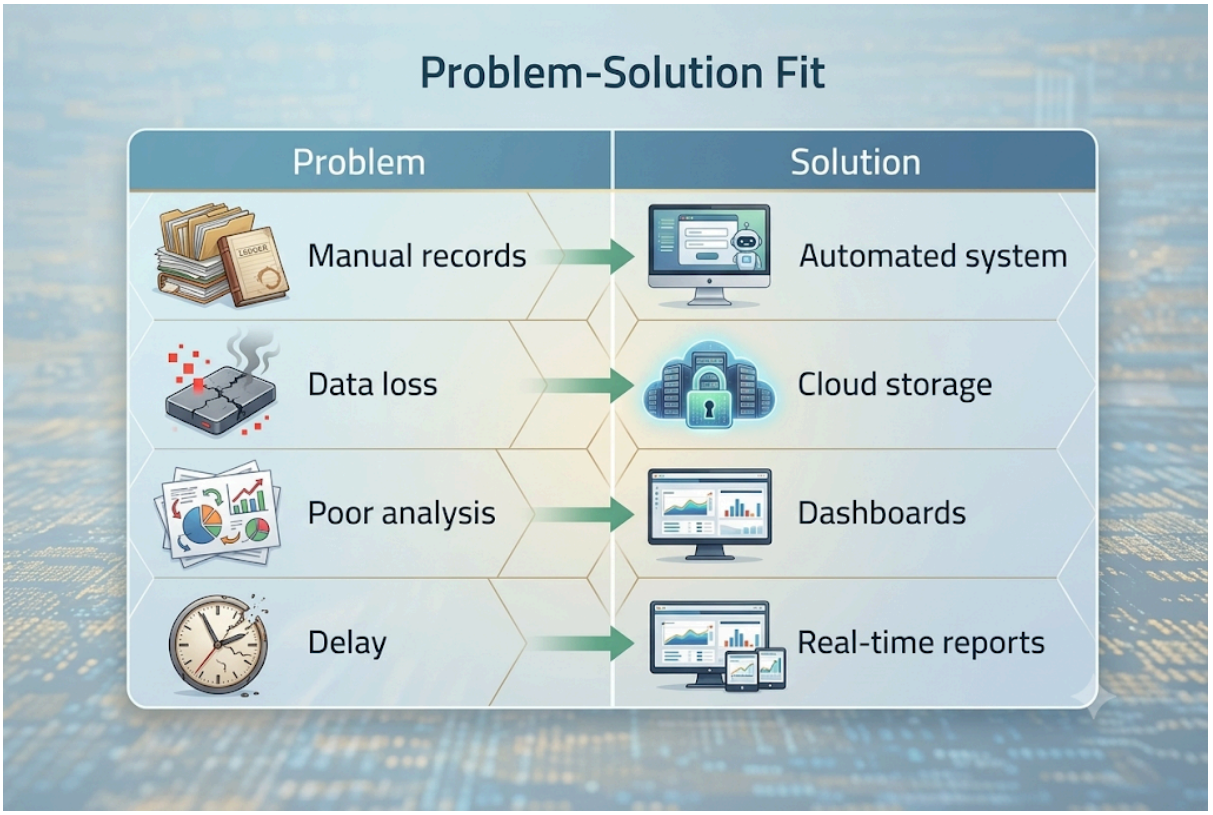
4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

Traditional cricket performance management relies heavily on manual records, which are time-consuming and error-prone. CRICKSTATS solves this by automating data entry and management through Salesforce.

- Data loss is prevented using secure cloud storage and automatic backups.
- Poor analysis is addressed through interactive dashboards and visual reports.
- Delays in reporting are eliminated with real-time reporting features.

Thus, CRICKSTATS bridges the gap between outdated manual processes and modern cloud-based solutions.



4.2 Proposed Solution

CRICKSTATS provides:

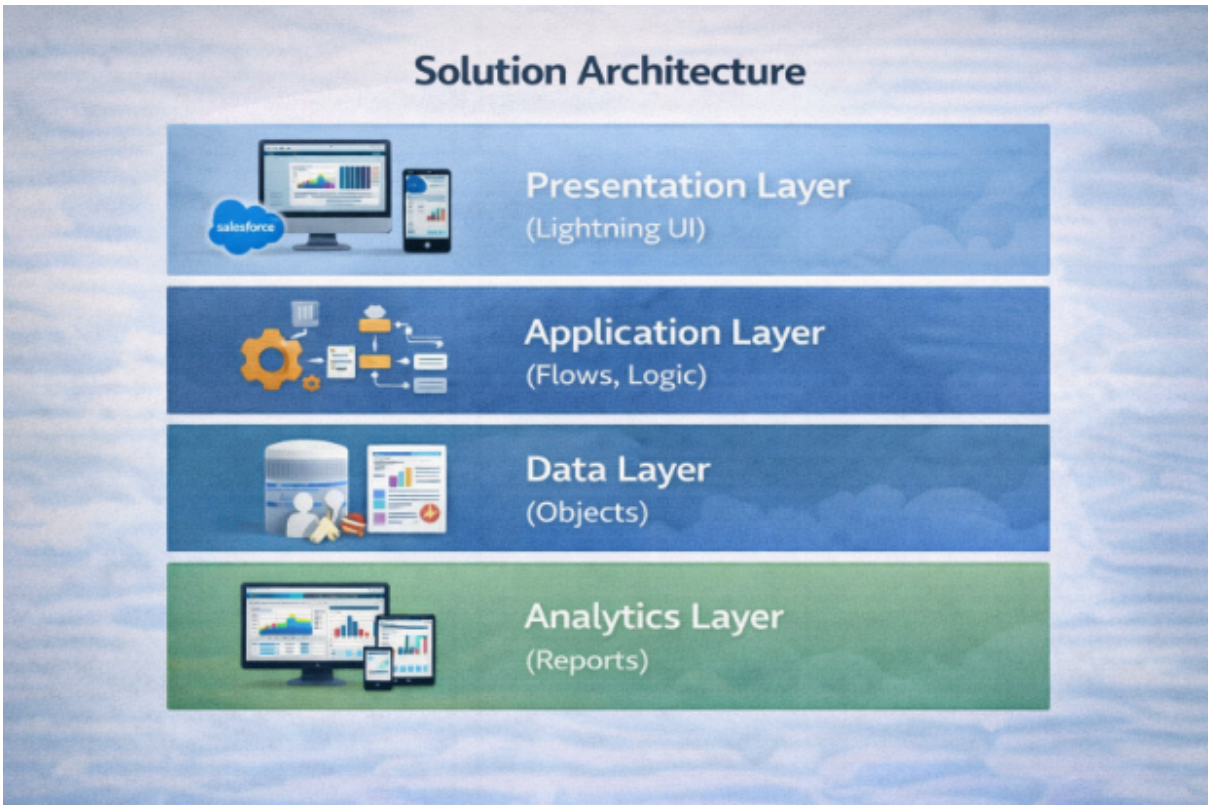
- Custom objects (Player, Match Performance)
- Master-detail and lookup relationships
- Roll-up summary fields
- Lightning App
- Automated flows
- Interactive dashboards

Key Features:

- Centralized database
- Live statistics
- Secure access
- Scalable design
- Easy reporting

4.3 Solution Architecture

Layer	Description
Presentation Layer	Lightning UI
Application Layer	Flows and Logic
Data Layer	Salesforce Objects
Analytics Layer	Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning – Detailed Explanation

What is Project Planning?

Project planning is the process of defining how the project will be executed, monitored, and delivered. It involves breaking down the work into smaller tasks, estimating effort, assigning responsibilities, and setting timelines.

For CRICKSTATS, I have planned the entire project using Agile methodology because it allows flexibility, continuous feedback, and incremental delivery.

My Planning Approach

I have planned the CRICKSTATS project based on three core principles:

1. Agile-based execution
2. Object creation → automation → reporting
3. Sprint-wise development

Let me explain each one in detail.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS system.	1	High	Member 1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Player and Match Performance objects so that player and match data can be stored systematically.	3	High	Member 2
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage records within the application.	3	High	Member 3
Sprint-2	UI Design	USN-4	As a coach, I want a dedicated Lightning App so that all performance tracking features are accessible from one dashboard.	3	High	Member 4
Sprint-2	Data Structure	USN-5	As an admin, I want to create player and match fields so that	5	Medium	Member 1

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			complete performance details can be captured accurately.			
Sprint-2	Data Structure	USN-6	As a user, I want customized page layouts so that only relevant fields are displayed during data entry.	5	High	Member 2
Sprint-2	Automation	USN-7	As a system, I want to validate player and match inputs so that incorrect data entries are prevented.	3	High	Member 4
Sprint-3	Automation	USN-8	As a system, I want automated flows to update player statistics so that performance data remains accurate.	5	High	Member 1

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. PROJECT DEVELOPMENT PHASE – Implementation

Project Flow (All Completed ✅)

I have successfully completed all seven milestones of the CRICKSTATS implementation:

- Milestone 1: Developer Account Creation — Done ✅
- Milestone 2: Object Creation — Done ✅
- Milestone 3: Tabs Creation — Done ✅
- Milestone 4: Lightning App Creation — Done ✅
- Milestone 5: Fields & Relationships — Done ✅
- Milestone 6: Roll-up Summary — Done ✅

- Milestone 7: Flows — Done ✅

Milestone 1 – Salesforce Developer Account Creation

What I Did

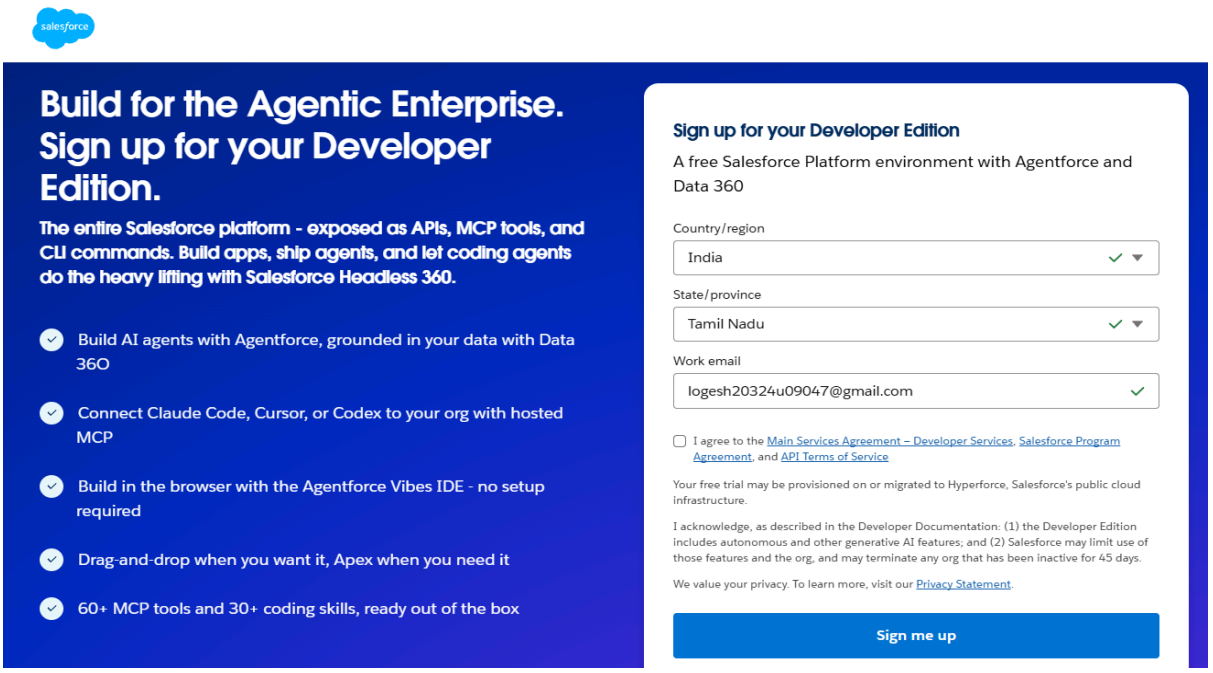
I created a Salesforce Developer Edition account to build and deploy the CRICKSTATS application. The Developer Edition provides a full-featured sandbox environment with API access, unlimited custom objects, and Lightning Experience enabled by default.

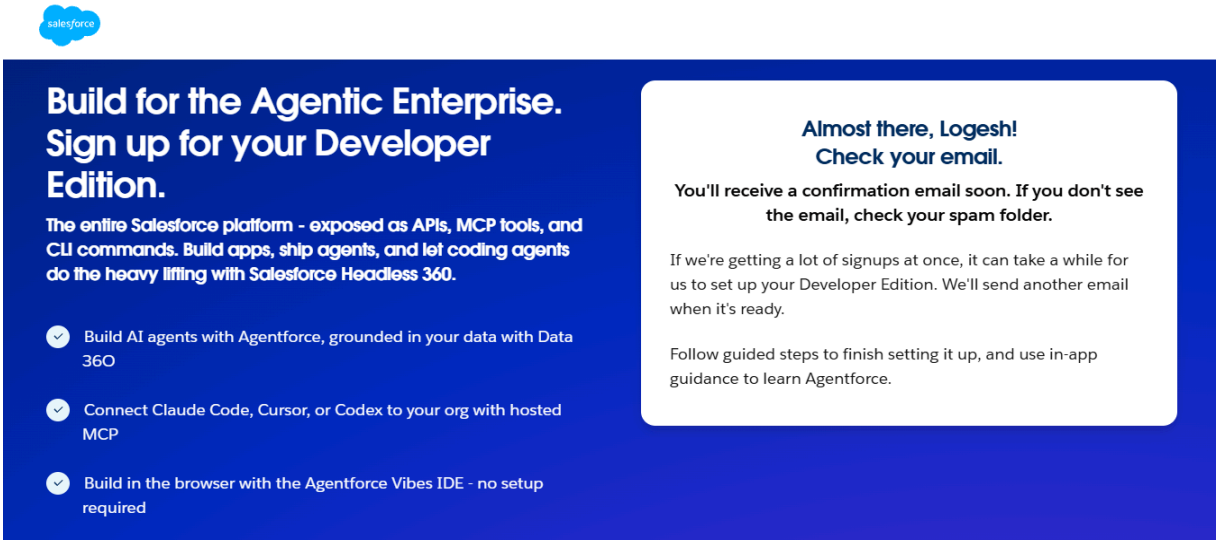
I created a Salesforce Developer Edition account to build and deploy the CRICKSTATS application. The Developer Edition provides a full-featured sandbox environment with:

- API access
- Unlimited custom objects
- Lightning Experience enabled by default
- Free forever (no cost)
- Safe environment to test without affecting production data

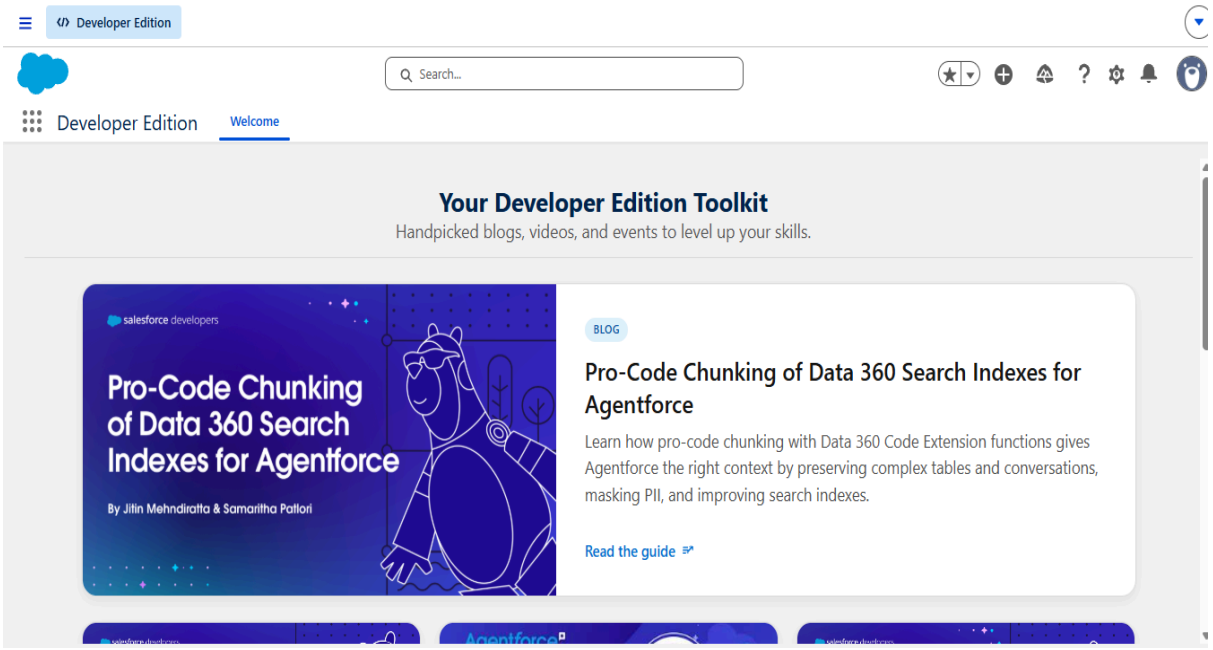
Step-by-Step Implementation

1. I navigated to `https://developer.salesforce.com/signup`
2. I filled in the signup form with the following details:
    - First Name & Last Name: My name
    - Email: My active email address
    - Role: Developer
    - Company: My college name
    - Country: India
    - Postal Code: My PIN code
    - Username: `username@organization.com` (this is a unique identifier, not a real email)
3. I clicked Sign Me Up
4. I opened my email inbox and clicked the Verify Account link (arrived within 5–10 minutes)



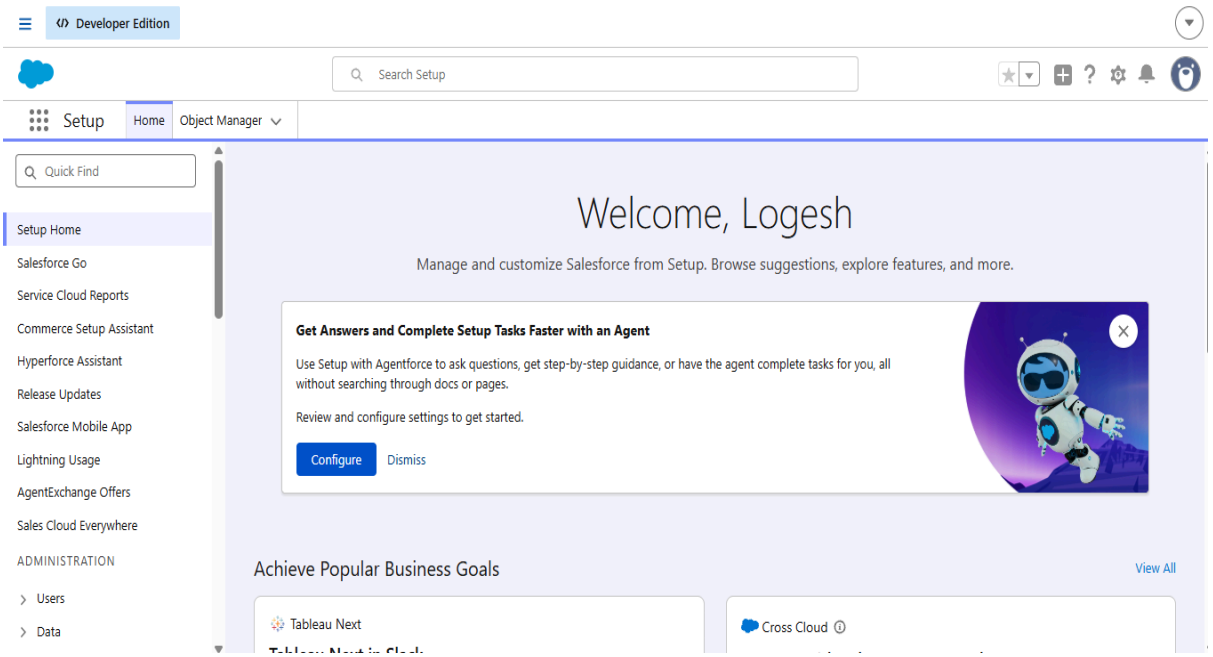


5. I set a strong password and answered a security question
6. I was redirected to the Salesforce Setup page, confirming successful account creation



Why This Step Matters

The Developer Edition is the foundation of the entire project. Without this, I cannot create custom objects, build flows, or deploy the Lightning App. It also provides a safe environment to test and iterate without affecting production data.



Milestone 2 – Object Creation

What I Did

I created two custom objects — Player and Match Performance — to store cricket-related data in a structured and relational manner.

Step-by-Step Implementation

Player Object:

1. I went to Setup → Object Manager → Create → Custom Object
2. I entered the following details:
    - Label: Player
    - Plural Label: Players
    - Record Name: Player Name
    - Data Type: Text
3. Under Optional Features, I checked:
    - Allow Reports
    - Allow Activities
    - Track Field History
4. Under Search Status, I checked Allow Search
5. I clicked Save

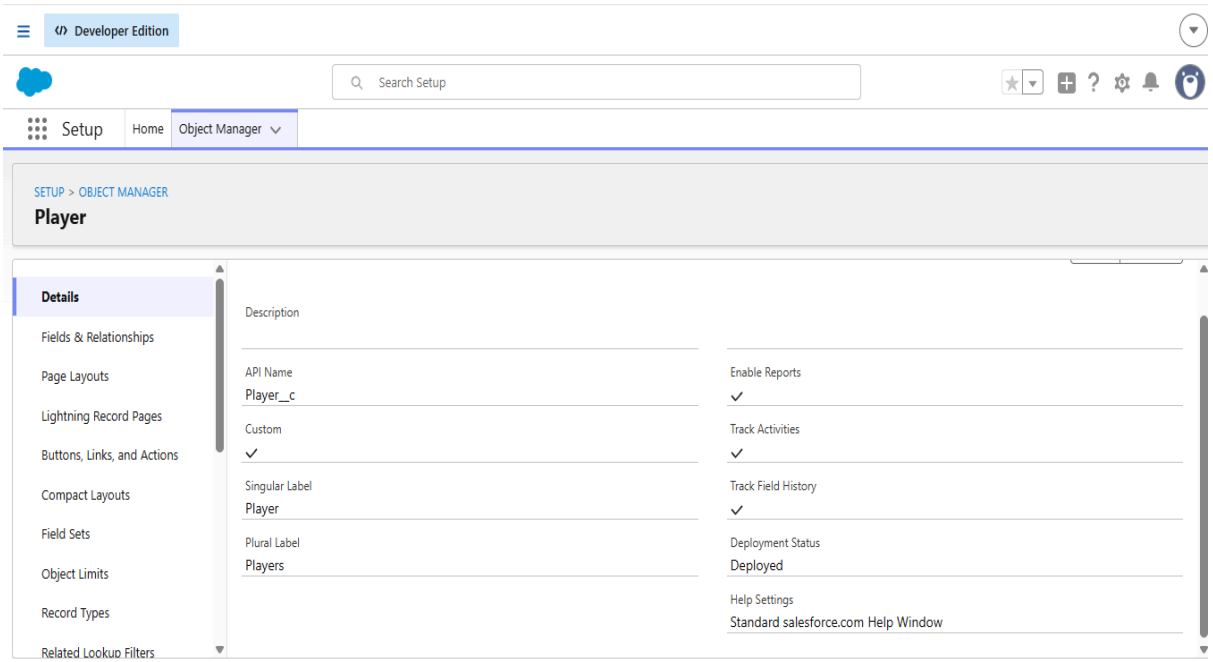
Why This Step Matters

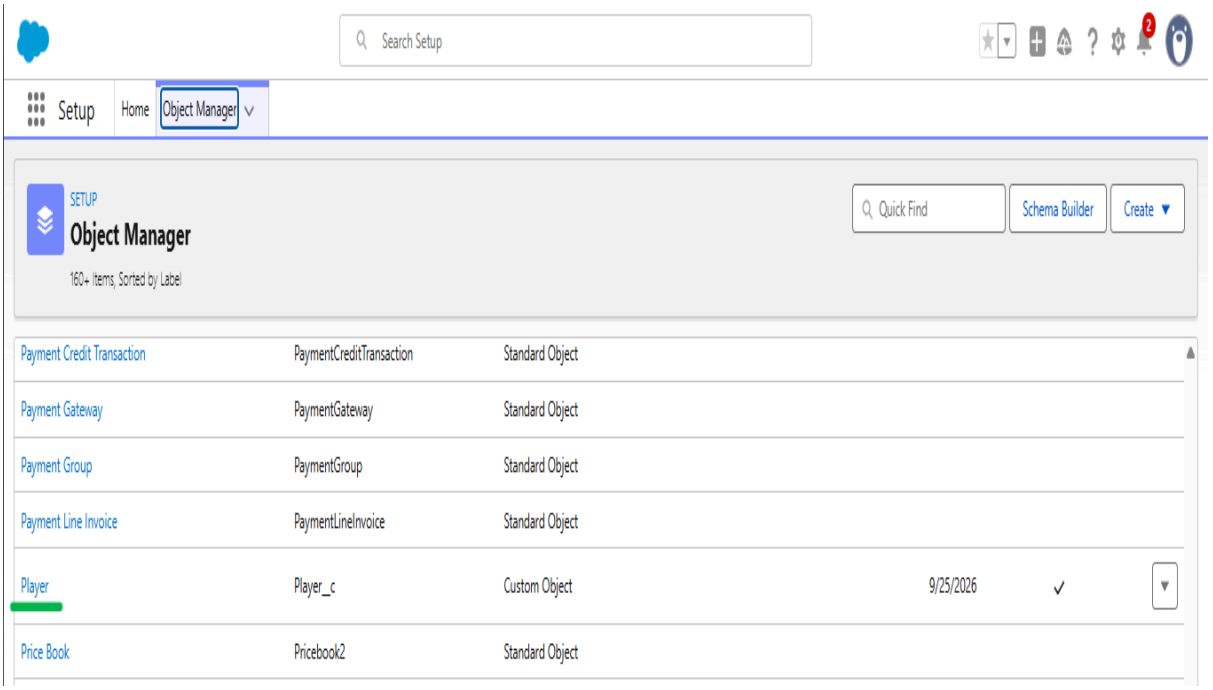
Custom objects are the backbone of any Salesforce application. By separating Player (master data) and Match Performance (transactional data), I ensured a clean data model.

Technical Reasoning:

- Allow Reports — enables reporting on this object from the start
- Allow Activities — allows tasks and events to be linked to records
- Track Field History — enables audit trail for changes
- Allow Search — makes records searchable via global search

Enabling these from the start means I can audit changes and build analytics later without rework.

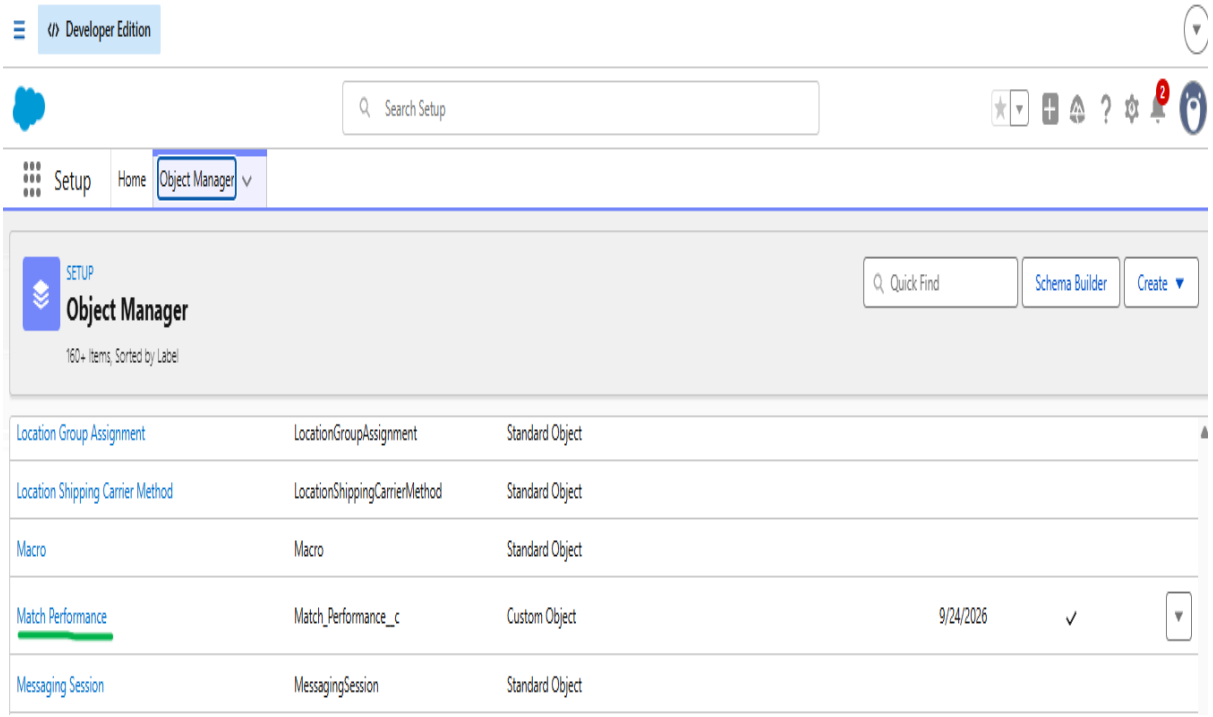




Match Performance Object:

1. I repeated the same navigation: Setup → Object Manager → Create → Custom Object
2. I entered:
    - Label: Match Performance
    - Plural Label: Match Performances

    - Record Name: Match Performance Name
    - Data Type: Text
3. I enabled the same optional features and search settings
4. I clicked Save



Why This Step Matters

Custom objects are the backbone of any Salesforce application. By separating Player (master data) and Match Performance (transactional data), I ensured a clean data model. Enabling Allow Reports and Track Field History from the start means I can audit changes and build analytics later without rework.

[Insert Image: Object Manager – Player and Match Performance Objects Created]

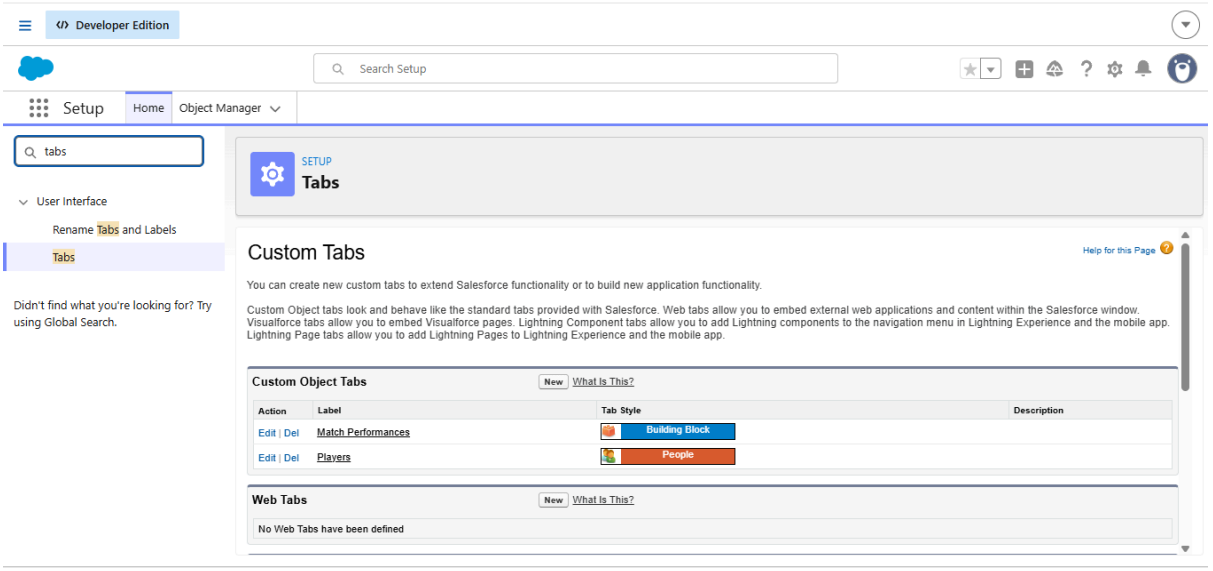
Milestone 3 – Tabs Creation

What I Did

I created custom tabs for both objects so users can easily navigate and manage records within the CRICKSTATS app.

Step-by-Step Implementation

1. I went to Setup → Quick Find → typed Tabs → clicked Tabs
2. Under Custom Object Tabs, I clicked New
3. I selected Player as the object
4. I chose a tab style (cricket-themed icon)
5. I clicked Next (Add to Profiles page — kept default)
6. I clicked Next (Add to Custom App — unchecked "Include Tab")
7. I ensured Append tab to users' existing personal customizations was checked
8. I clicked Save
9. I repeated the same steps for Match Performances



Why This Step Matters

Tabs provide direct navigation to objects. Without tabs, users would have to search through Setup or App Launcher every time. Tabs improve usability and reduce the learning curve — a key non-functional requirement (NFR-1: Usability).

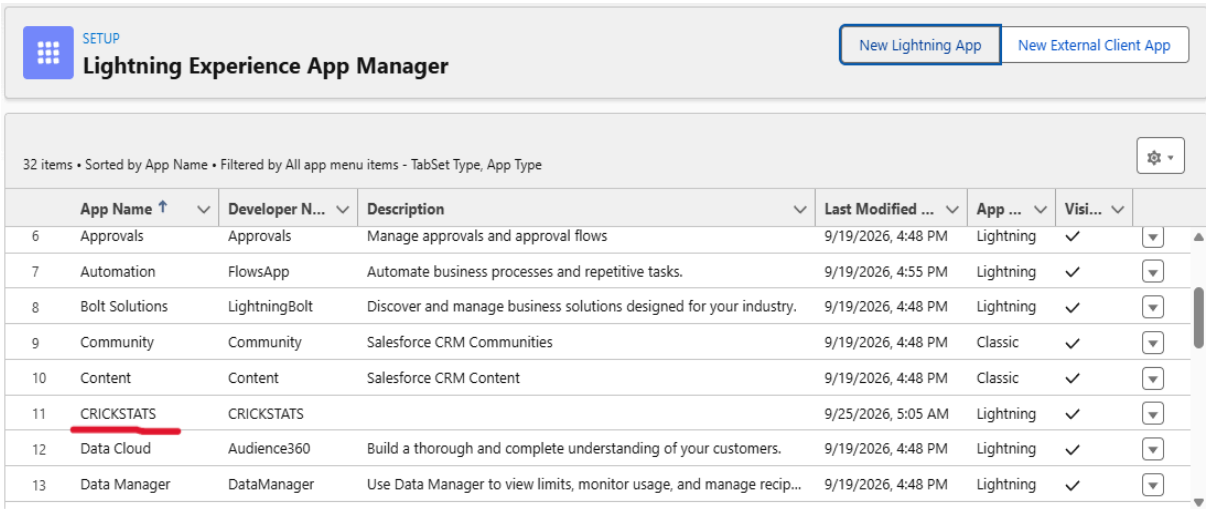
Milestone 4 – Lightning App Creation

What I Did

I created a custom Lightning App called CRICKSTATS that bundles all objects, reports, and dashboards into a single, unified interface.

Step-by-Step Implementation

1. I went to Setup → Quick Find → App Manager → New Lightning App
2. App Details:
    - App Name: CRICKSTATS
    - Developer Name: Auto-populated
    - Primary Color: Default
3. App Options:
    - Navigation Style: Standard Navigation
    - Setup Experience: Setup
    - Supported Form Factors: Desktop and Phone
4. Utility Items: Kept default alignment
5. Navigation Items: I selected:
    - Players
    - Match Performances
    - Reports
    - Dashboards
6. User Profiles: I selected System Administrator
7. I clicked Save & Finish



Why This Step Matters

The Lightning App is the user-facing container. By grouping objects, reports, and dashboards into one app, users don't need to switch between multiple tabs or apps. This directly supports the goal of a centralized system and improves productivity.

Milestone 5 – Fields & Relationships

What I Did

I created all required custom fields on both objects to capture complete player and match performance data.

Step-by-Step Implementation

Player Object Fields:

1. I went to Object Manager → Player → Fields & Relationships → New
2. I created the following fields:

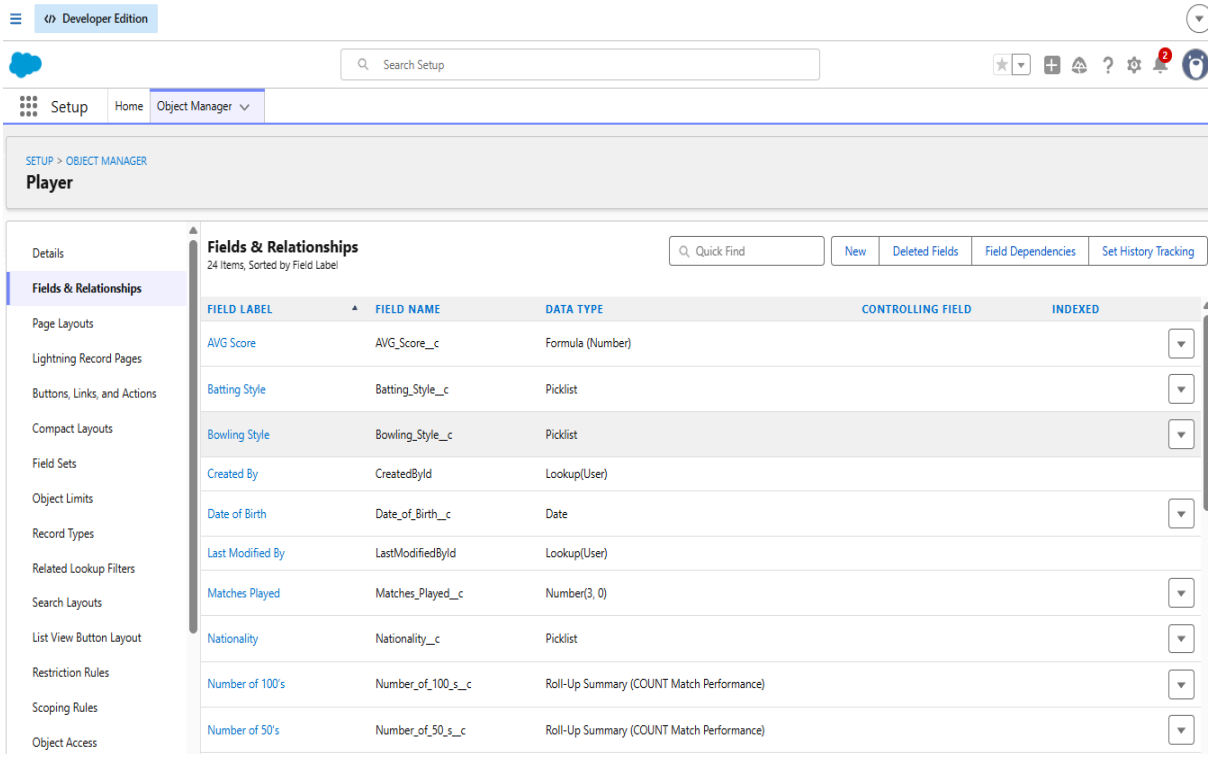
Field Label	Data Type	Details
Player Name	Text	Length: 50
Player Number	Text	Length: 3
Profile Picture	Rich Text Area	—
Date of Birth	Date	—

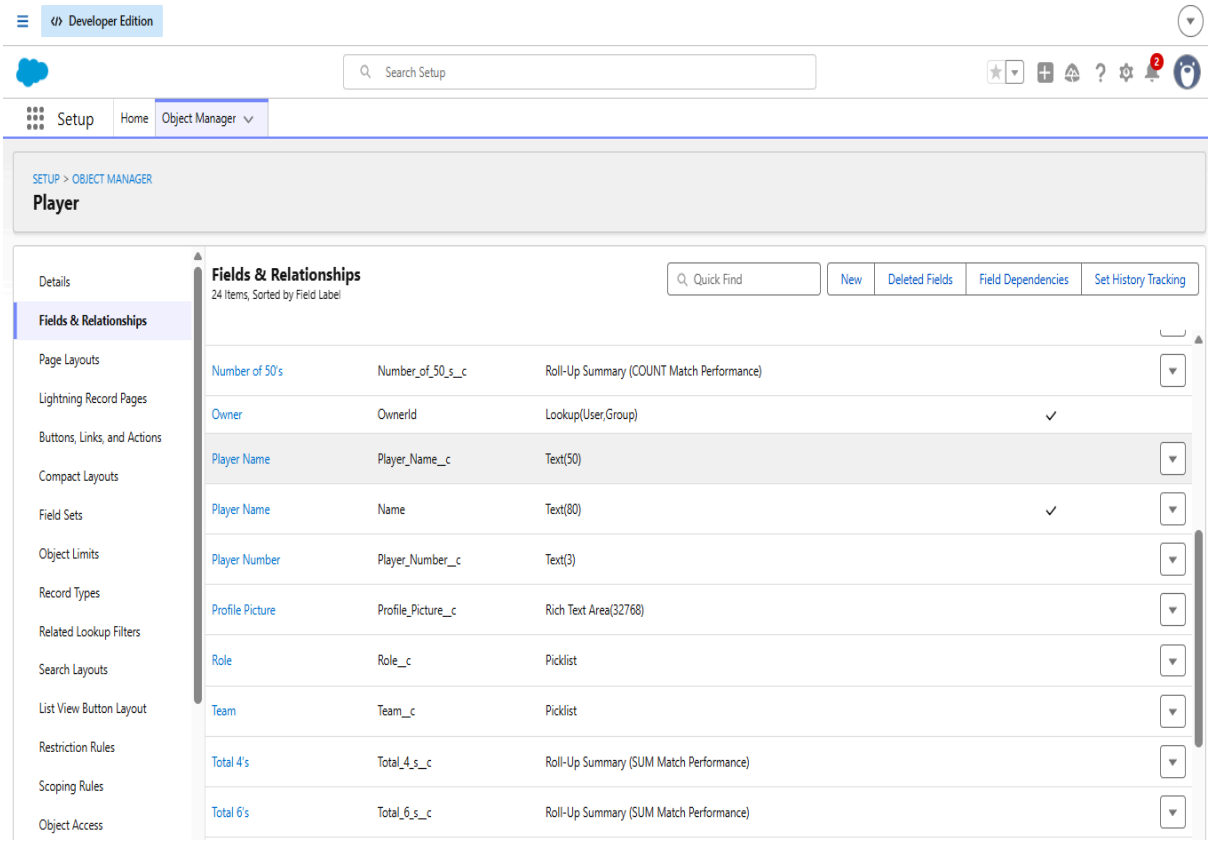
Role	Picklist	Batsman, Bowler, Batsman & Wicket Keeper, All Rounder
Batting Style	Picklist	Right-Handed, Left-Handed
Bowling Style	Picklist	Right-Arm Fast, Left-Arm Fast, Right-Arm Medium, Left-Arm Medium, Right-Arm Spin, Left-Arm Spin
Nationality	Picklist	Australia, Bangladesh, England, India, New Zealand, Pakistan, South Africa, Sri Lanka, West Indies, Afghanistan, Zimbabwe, Ireland, Scotland, Netherlands, UAE, Nepal, USA
Team	Picklist	CSK, DC, KKR, MI, PBKS, RR, RCB, SRH, GT, LSG
AVG Score	Formula (Number)	`IF(Matches_Played__c > 0, Total_Runs__c / Matches_Played__c, 0)`
Match Performance	Lookup Relationship	Related To: Match Performance

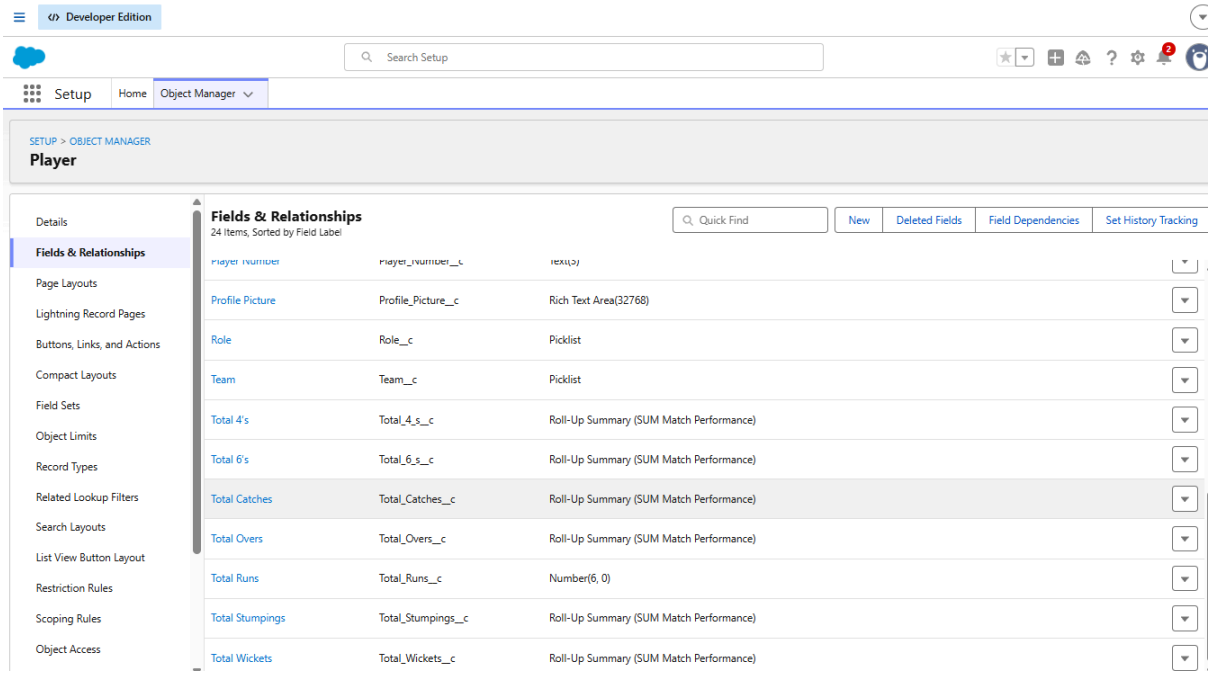
Match Performance Object Fields:

Field Label	Data Type	Details
Match Performance Name	Text	Length: 80
Player of the Match	Checkbox	—
Match Result	Picklist	WON, LOST, DRAW
Runs Scored	Number	—
Balls Faced	Number	—
Fours	Number	—
Sixes	Number	—
Overs Bowled	Number	—
Wickets Taken	Number	—
Runs Conceded	Number	—
Maiden Overs	Number	—
Stumpings	Number	—

Catches	Number	—
Player	Master-Detail Relationship	Related To: Player







Why This Step Matters

Fields define the data structure. The formula field (AVG Score) dynamically calculates batting average without manual input. The lookup relationship on Player links to Match Performance, while the master-detail relationship on Match

Performance ensures that match records cannot exist without a parent Player — enforcing referential integrity.

Milestone 6 – Roll-up Summary

What I Did

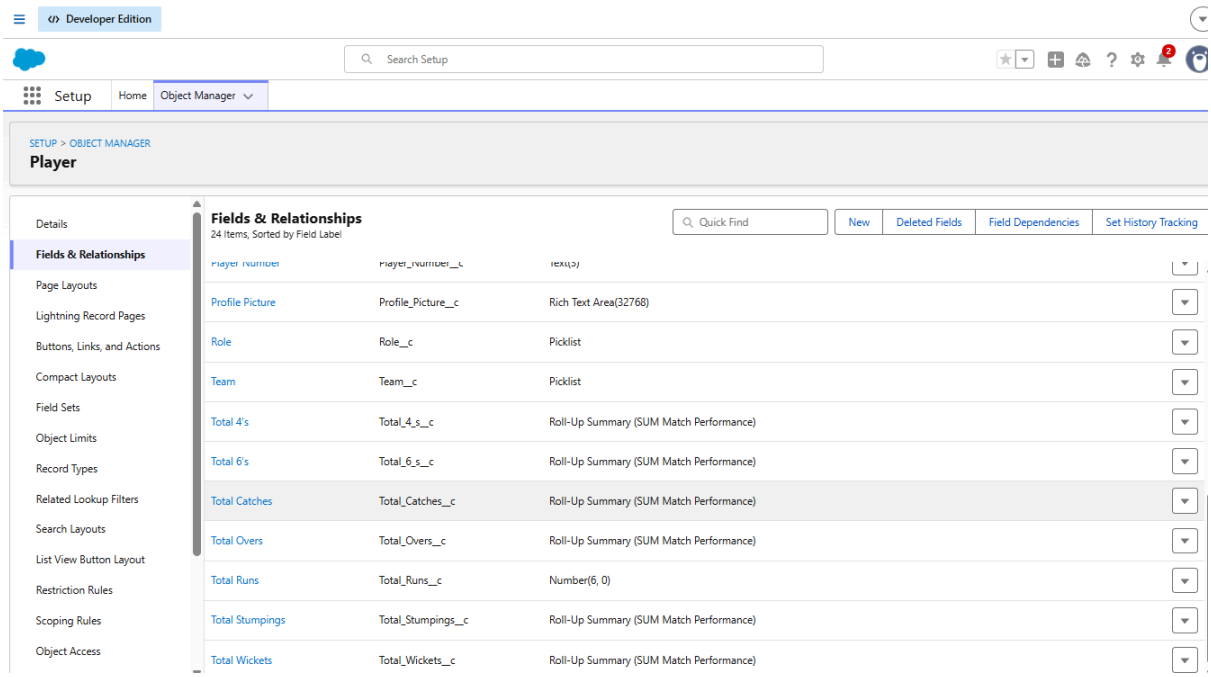
I created roll-up summary fields on the Player object to automatically aggregate match performance data.

Step-by-Step Implementation

1. I went to Object Manager → Player → Fields & Relationships → New
2. I selected Roll-up Summary as the data type
3. I configured the following roll-up fields:

Field Label	Roll-up Type	Field to Aggregate	Filter Criteria
Matches Played	COUNT	—	All records
Number of 50's	COUNT	—	Only records meeting criteria
Number of 100's	COUNT	—	Only records meeting criteria
Total 4's	SUM	Fours	All records
Total 6's	SUM	Sixes	All records

Total Runs	SUM	Runs Scored	All records
Total Overs	SUM	Overs Bowled	All records
Total Wickets	SUM	Wickets Taken	All records
Total Stumpings	SUM	Stumpings	All records
Total Catches	SUM	Catches	All records



Why This Step Matters

Roll-up summary fields are a declarative automation tool in Salesforce. They eliminate the need for manual calculation or Apex triggers. Since they only work on the parent object in a master-detail relationship, the Player object (parent)

automatically aggregates data from Match Performance (child). This ensures real-time accuracy and reduces data redundancy.

[Insert Image: Roll-up Summary Field Configuration – Matches Played]

Milestone 7 – Flows

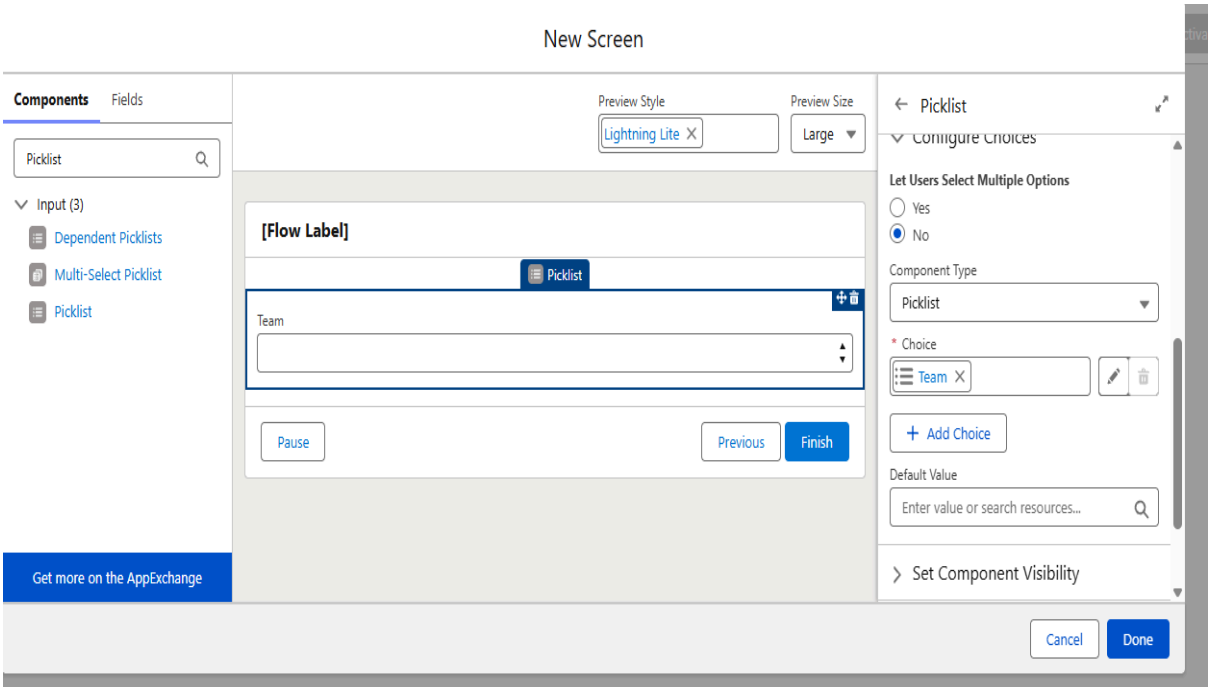
What I Did

I created a Screen Flow called Player Flow that allows users to select a team and view a filtered data table of players with their performance metrics.

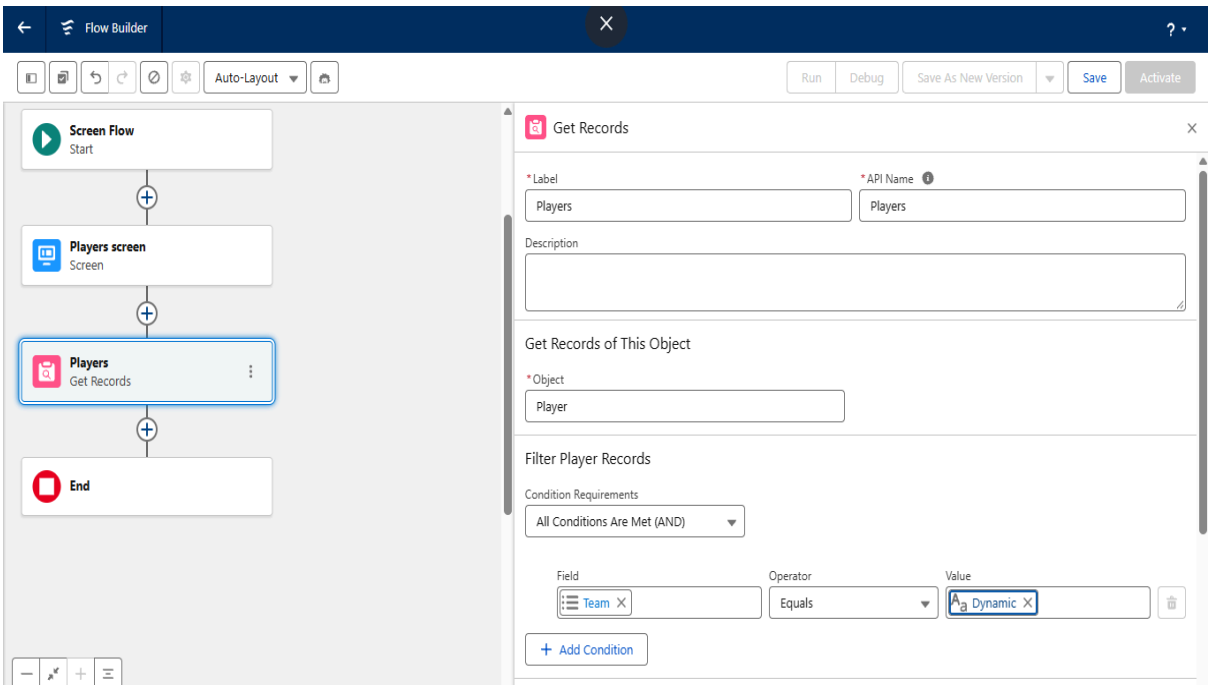
Step-by-Step Implementation

Activity 1: Screen Flow Creation

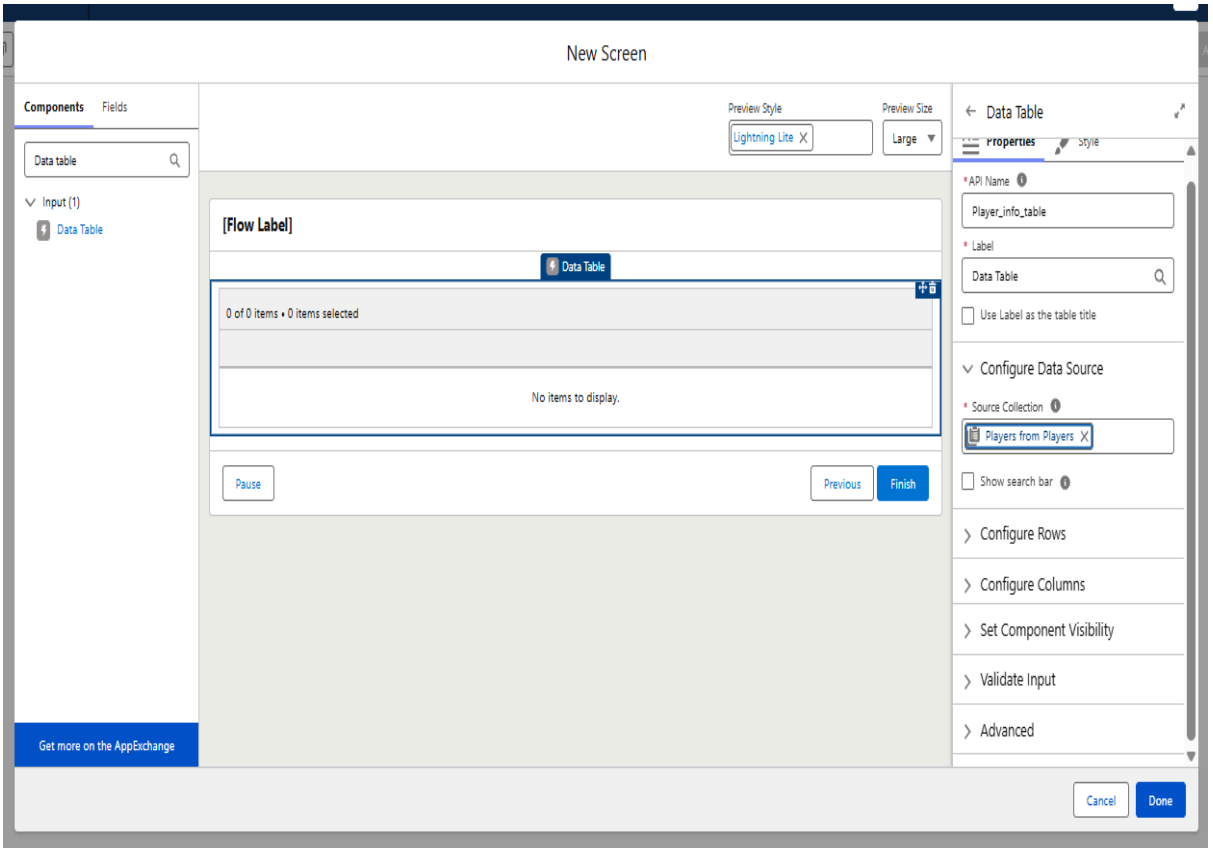
1. I went to Setup → Quick Find → Flows → New Flow
2. I selected Screen Flow and clicked Create
3. I clicked the (+) icon and added a Screen Component
    - Label: Players_screen
    - API Name: Players_screen
4. I added a Picklist component to the screen:
    - Label: Team
    - API Name: Team
    - Data Type: Text
    - Choice: New Choice Resource → Picklist Choice Set
    - API Name: Team
    - Object: Player
    - Data Type: Picklist
    - Field: Team



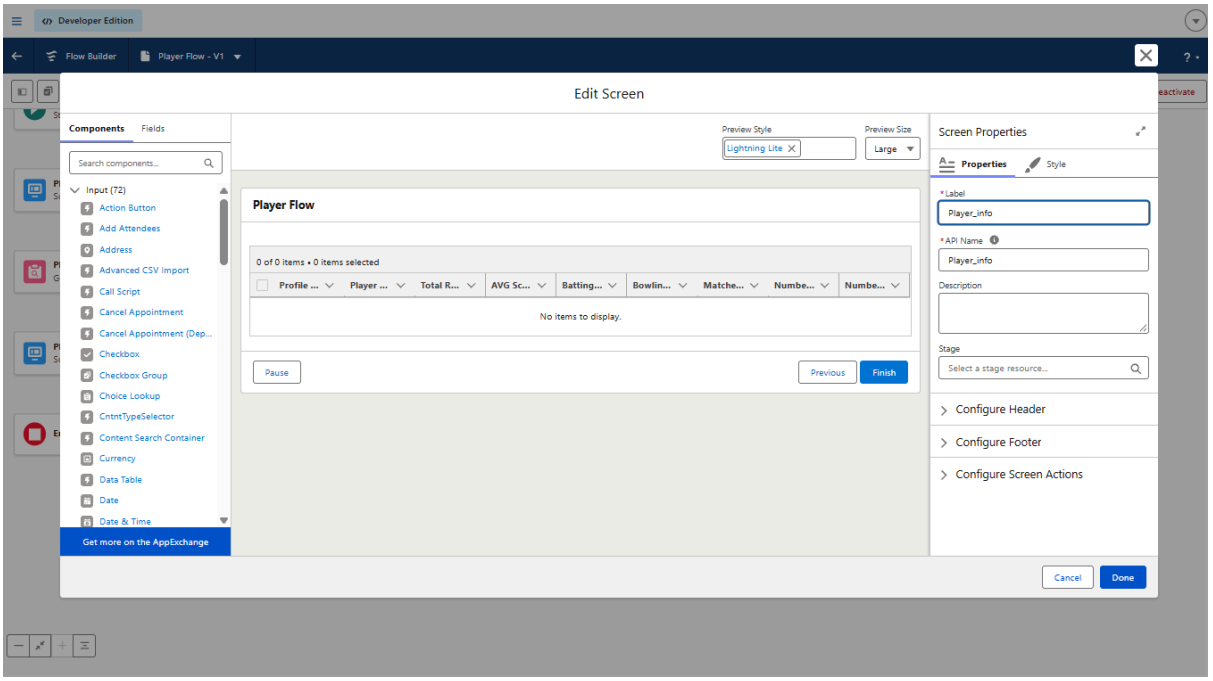
5. I clicked (+) and added a Get Records component:
    - Label: Players
    - API Name: Players
    - Object: Player
    - Filter: Field = Team, Operator = Equals, Value = Dynamic Variable
    - New Resource: Variable (API Name: dynamic, Data Type: Text, Default Value: Screen component → Team Picklist)



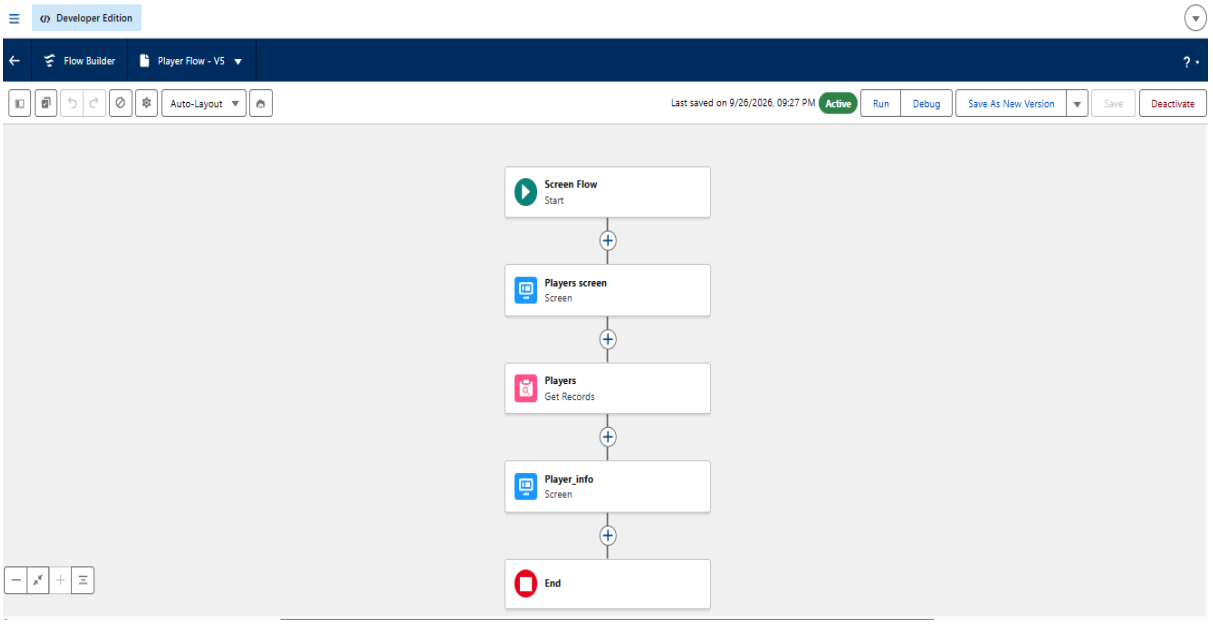
6. I clicked (+) and added another Screen Component:
    - Label: Player_info
    - API Name: Player_info
    - Component Type: Data Table
    - API Name: Player_info_table
    - Source Collection: Players from Players (Player__c)



7. In Configure Columns, I added:
    - Profile Picture
    - Player Name
    - Total Runs
    - AVG Score
    - Batting Style
    - Bowling Style
    - Matches Played
    - Number of 50's
    - Number of 100's

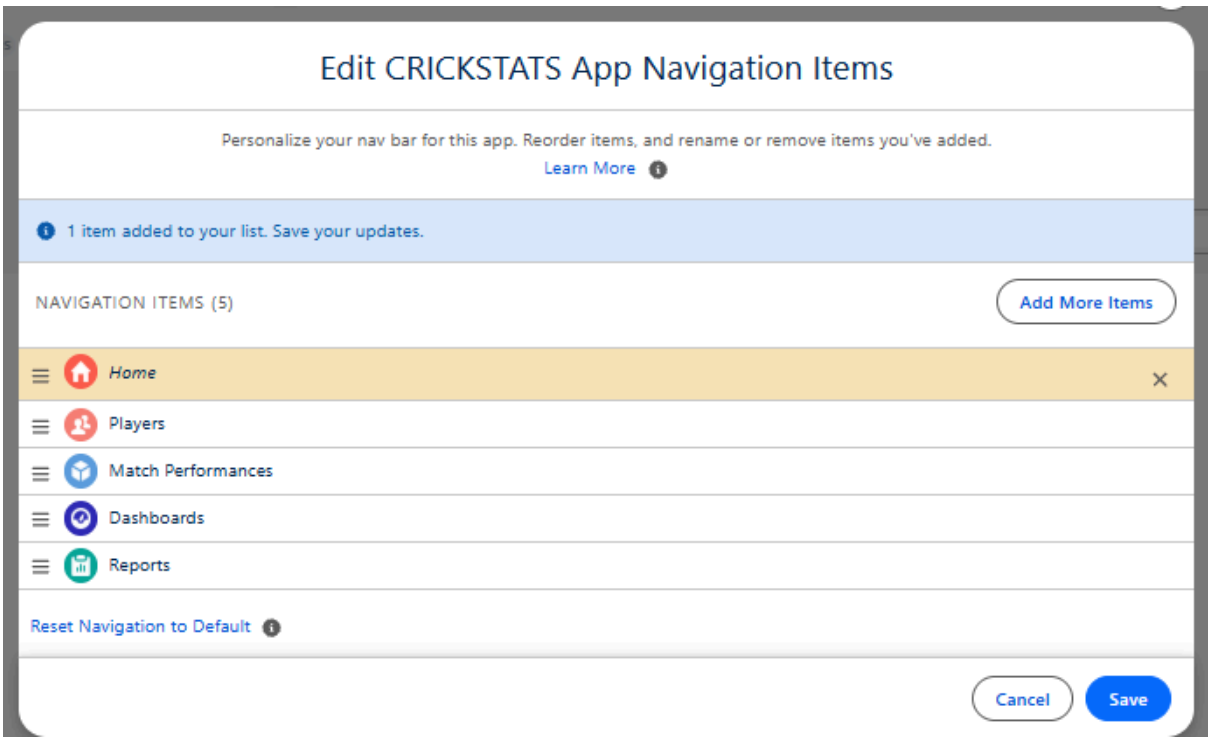


8. I clicked Done → Save
    - Flow Label: Player Flow
    - Flow API Name: Player_Flow



Activity 2: Adding Flow to Home Page

1. I opened the CRICKSTATS app → clicked the Personalize your nav bar icon (pencil icon)
2. I added Home Page and clicked Save



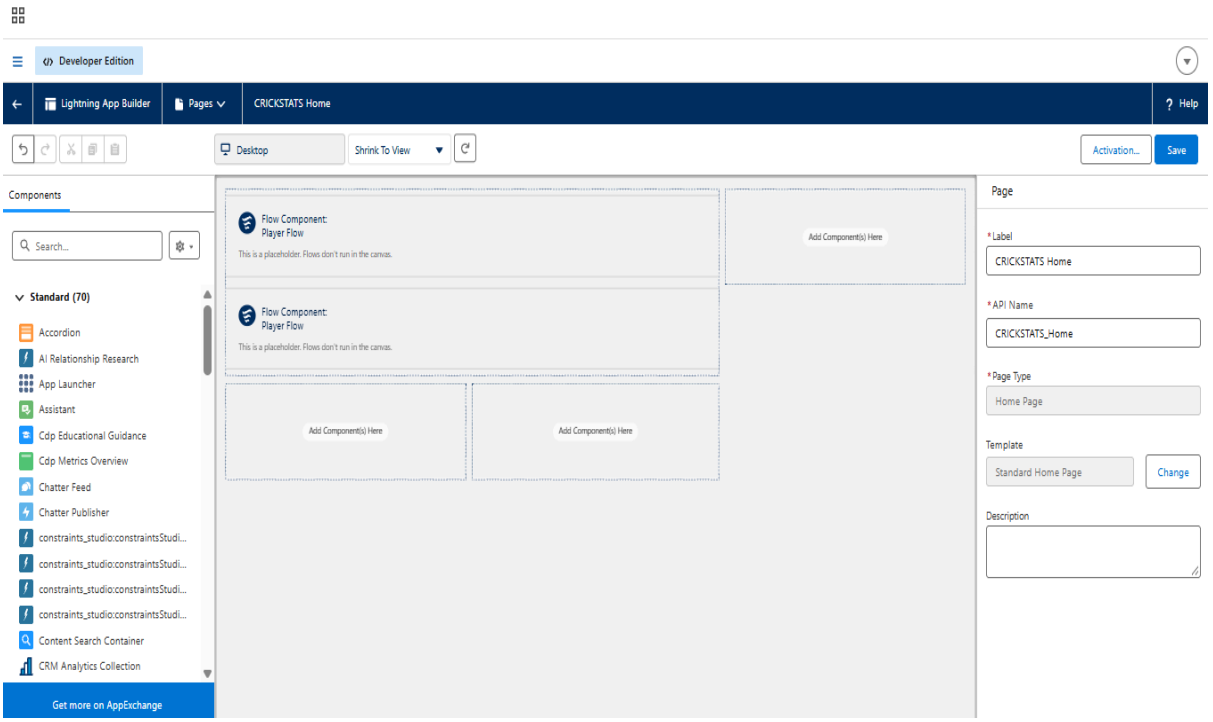
3. I opened Lightning App Builder for the Home Page
4. I removed all other components and added the Flow component
5. I selected Players Flow from the Flow dropdown
6. I clicked Save and Activation
7. I clicked App Default → Assign to Apps
8. I selected CRICKSTATS App → Next → Save

Why This Step Matters

Flows are the automation engine of Salesforce. The Screen Flow I built provides a user-friendly interface for coaches and managers to:

- Select a team from a dropdown
- Instantly view a filtered list of players
- See key performance metrics in a data table

This eliminates the need for manual filtering and makes the system interactive and real-time. By embedding the flow on the Home Page, users see it immediately upon opening the app — improving adoption and usability.

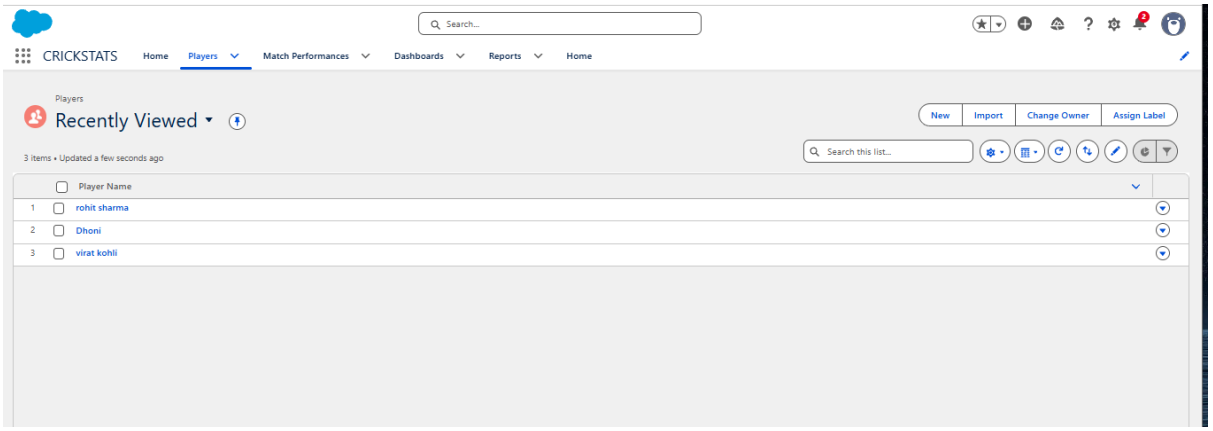


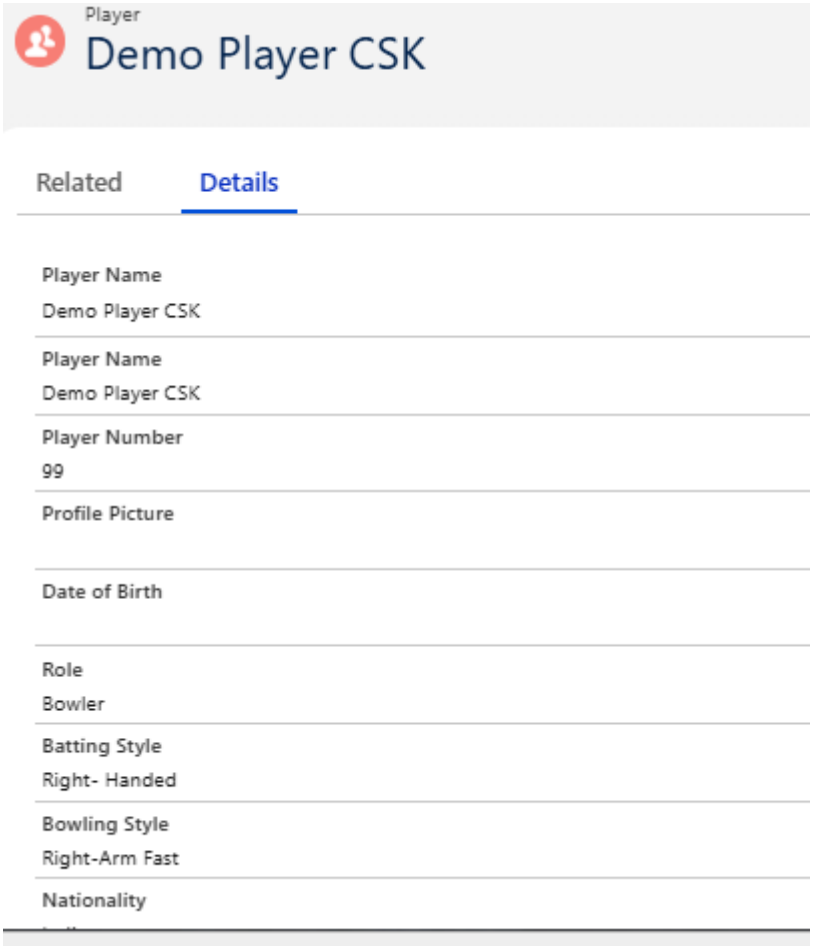
Why This Step Matters

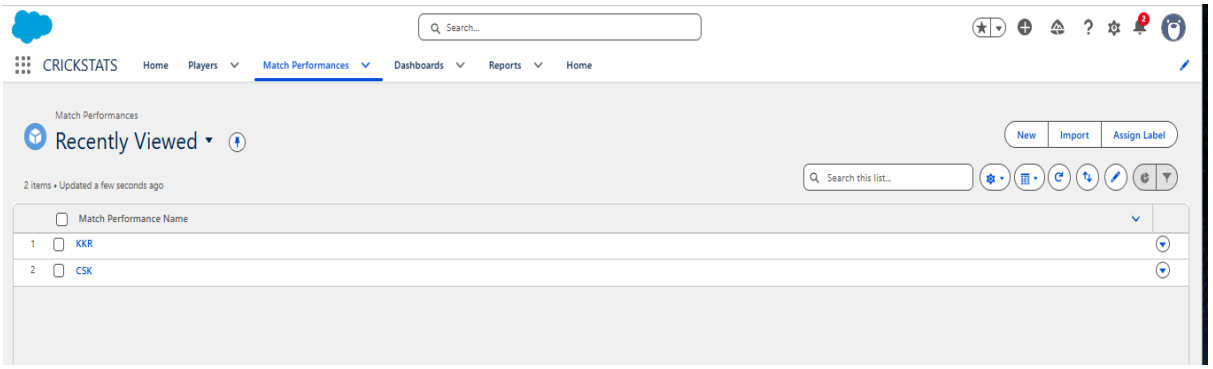
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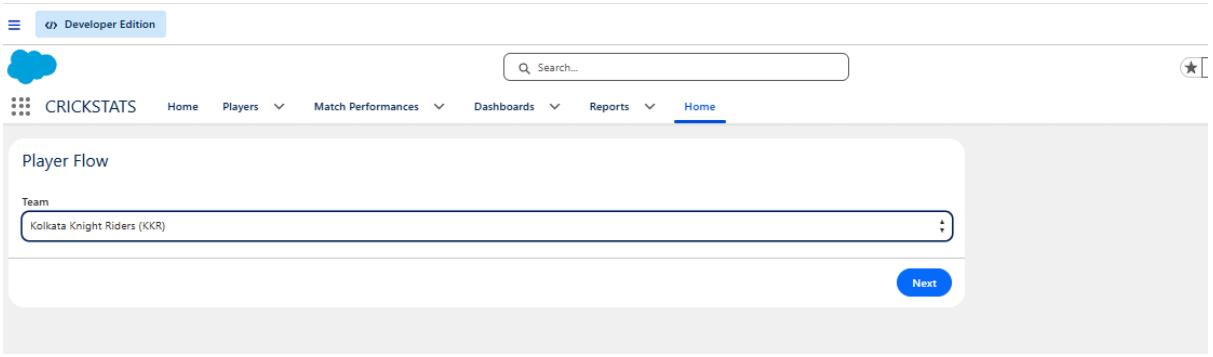
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Search...
CRICKSTATS
Home
Players
Match Performances
Dashboards
Reports
Home
Players
Recently Viewed
New
Import
Change Owner
Assign Label
3 items • Updated a few seconds ago
Search this list...
Player Name
1 rohit sharma
2 Dhoni
3 virat kohli


Player
Demo Player CSK
Related
Details
Player Name
Demo Player CSK
Player Name
Demo Player CSK
Player Number
99
Profile Picture
Date of Birth
Role
Bowler
Batting Style
Right- Handed
Bowling Style
Right-Arm Fast
Nationality

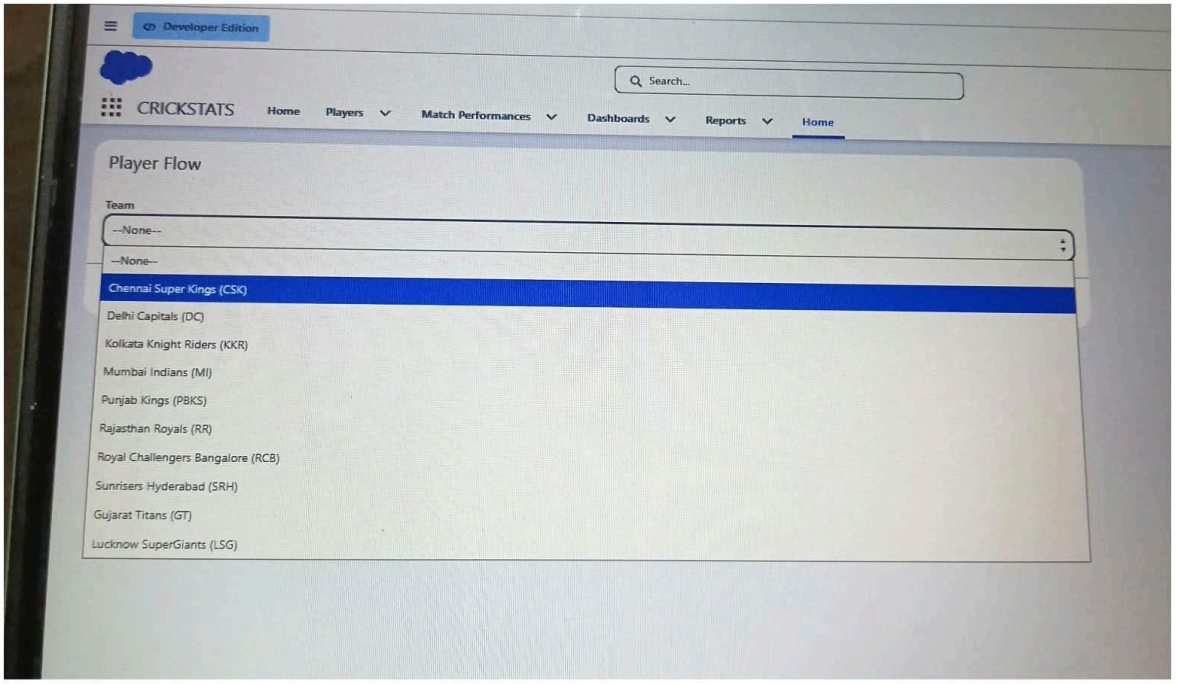

Search...
CRICKSTATS
Home
Players
Match Performances
Dashboards
Reports
Home
Match Performances
Recently Viewed
New
Import
Assign Label
2 items • Updated a few seconds ago
Search this list...
Match Performance Name
1 KKR
2 CSK

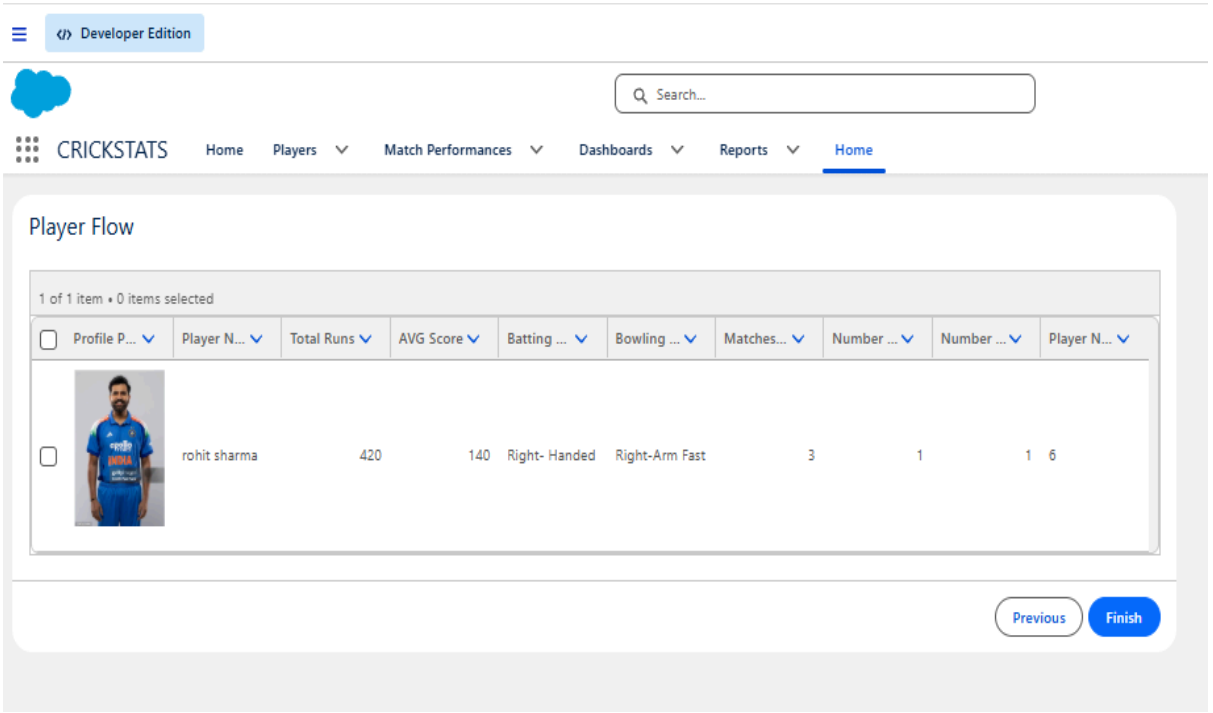


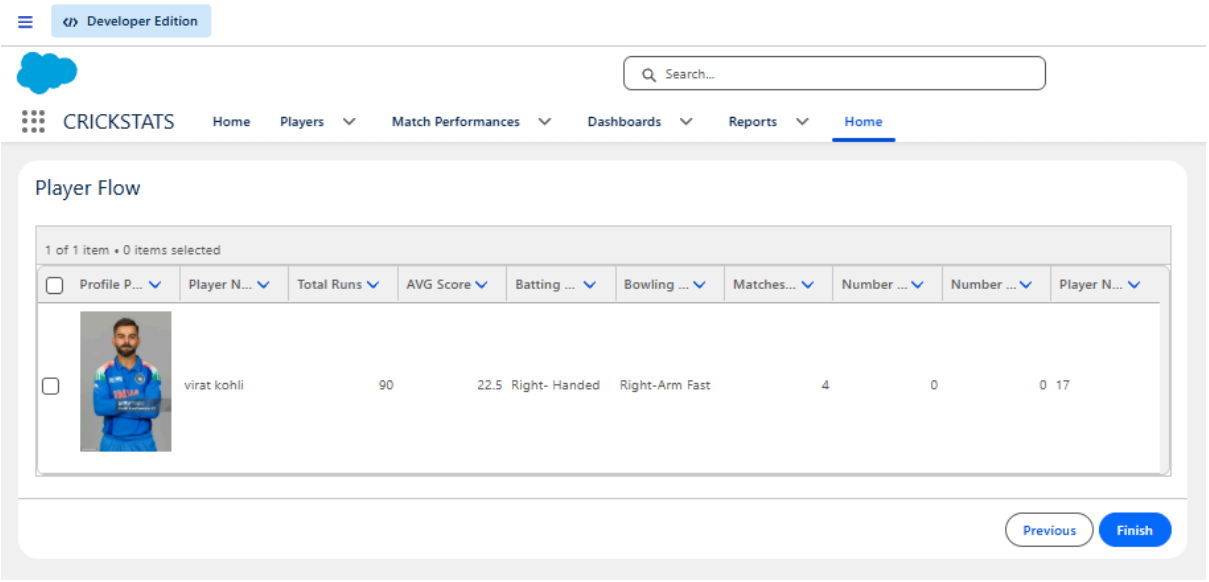
Summary of Implementation

Milestone	What I Completed	Status
1	Salesforce Developer Account	✅ Done
2	Player & Match Performance Objects	✅ Done
3	Custom Tabs	✅ Done
4	CRICKSTATS Lightning App	✅ Done
5	All Custom Fields & Relationships	✅ Done
6	Roll-up Summary Fields	✅ Done
7	Screen Flow & Home Page Integration	✅ Done

FINAL PROJECT OUTPUT SCREEN SHORT :







Technical Highlights of My Implementation

1. Data Integrity: Master-detail relationship ensures match records cannot exist without a player.
2. Real-Time Aggregation: Roll-up summary fields automatically calculate totals.
3. Dynamic Formula: AVG Score updates automatically based on matches played and total runs.
4. User-Friendly Automation: Screen Flow provides a guided, interactive experience.
5. Centralized Access: Lightning App bundles everything into one interface.
6. Scalability: Salesforce cloud infrastructure handles growing data volumes.

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from the Salesforce Lightning UI to verify:

- Record creation for Player and Match Performance objects
- Validation rules for mandatory fields and input constraints
- Record type behavior for Domestic and International matches
- Flow execution for automatic updating of player statistics

- Roll-up summary accuracy for runs, wickets, averages, and other metrics

Performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading. The application was tested under multiple user interactions to ensure stable performance without delays. Screenshots of reports and dashboards confirmed accurate data visualization, real-time updates, and reliable performance tracking.

Note: Enter sample records in Player and Match Performance objects to see results.

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized system
- Real-time analytics
- Secure platform
- Scalable design
- Reduced manual work

Disadvantages

- Requires Salesforce license
- Internet dependency
- Limited offline access

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9. CONCLUSION

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development.

CRICKSTATS demonstrates how Salesforce can be used to develop a structured application for managing cricket player performance data. The project uses custom objects for Player and Match Performance records, along with fields and relationships designed to connect player profiles with their match statistics.

The application's configured summary fields and formula support the calculation and display of selected performance metrics. A team-based screen flow provides a guided way to select a team and review matching player records. These elements demonstrate the use of Salesforce data modeling, declarative configuration, and automation in a practical application.

The project documentation describes the intended benefits as centralized data management, reduced manual work, and easier access to performance information. The usefulness of these benefits in practice depends on accurate data entry, appropriate configuration, and confirmation that the relevant features are working in the deployed Salesforce org.

10. FUTURE SCOPE

The project documentation identifies several possible extensions for future development. These are proposed enhancements and should not be considered implemented features unless they are built and tested.

- Mobile application integration
- AI-based performance prediction
- Video analysis integration
- Player ranking system
- Public statistics portal
- Multi-league support

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11. APPENDIX

- Salesforce Objects & Fields
- Manual match statistics
- Flow configurations

ACKNOWLEDGEMENT :

I also acknowledge the Salesforce learning environment and its learning resources, which helped me understand cloud-based application development, custom data modeling, automation, and reporting. These resources supported the configuration and development of the CRICKSTATS application.

Thank You